VAYUSETU: Jury QA Document

Comprehensive Answers to 20 Anticipated Jury Evaluation Questions

1. Strategic Core Questions

Q1: How does VAYUSETU differ from standard meteorological web dashboards?
Answer: Standard dashboards display raw, unlinked observations. VAYUSETU functions as a Digital Twin. It runs closed-loop Kalman filter data assimilation, linking live data with physical and AI predictors. Changing any state updates all dependent risk indicators, maps, and stakeholder advisories automatically in a mathematically consistent way.

Q2: What satellite and observation assets are actually used?
Answer: We utilize real IMD daily rainfall (0.25° grid) and temperature (1° grid) observations. We also integrate INSAT-3D Land Surface Temperature, Sea Surface Temperature, and Rainfall NetCDF grids, along with Bhuvan Land Use Land Cover (LULC) maps for runoff calculations.

Q3: What models comprise the AI Ensemble?
Answer: ConvLSTM (for gridded precipitation movements), Temporal Fusion Transformer (TFT for long-term multi-hazard forecasts), XGBoost (for residual corrections based on local elevation), and PINN (Physics-Informed Neural Network to ensure energy-water conservation bounds).

Q4: How is the Climate Resilience Index (CRI) score computed?
Answer: It uses a weighted aggregation: 35% Flood Risk + 35% Heatwave Risk + 15% Drought Risk + 15% Water Stress. The score is classified as SAFE (0-25), MODERATE (25-50), HIGH (50-75), or CRITICAL (75-100). This prevents contradictions where high individual hazards are masked as safe.

Q5: How is the 2D flood routing simulation calculated?
Answer: We solve the Saint-Venant shallow water equations using a finite-difference grid method. It simulates how extreme precipitation translates into 2D flood depth contours across the basin based on Bhuvan soil types and slope slopes.

2. Technical & Architecture Questions

Q6: What is the purpose of the Digital Twin Lineage Engine?
Answer: It provides absolute auditability. For every twin state generated, it logs the exact file checksums, satellite run IDs, operator signatures, and coverage scores, showing the exact heritage of the twin state.

Q7: How does the AI copilot answer questions dynamically?
Answer: The copilot service doesn't use hardcoded text. It parses the current active Twin State parameters (CRI, temperature, rainfall, soil moisture) and routes them into a rule-based inference flow that matches current states with scientific climate response guidelines.

Q8: Can this scale to a national level?
Answer: Yes. The database schema separates districts into nodes. The Docker configuration containerizes the FastAPI backend and Next.js frontend, allowing deployment on Kubernetes cluster nodes to handle high-concurrency requests.